Review Problems
AP Calculus
Sec 6.1 – 6.9

Name SOLUTIONS

These are sample questions from unit 6. This is not an exact review for the test, just practice problems.

Multiple Choice:

1. Let $f(x)$ be a continuous function. What is the value of $\int_3^{15} f(x)\,dx$ if it is known that $\int_1^5 f(3x)\,dx = 7$?

(A) $\frac{7}{3}$ (B) 7

$\boxed{\text{(C) } 21}$ (D) $\frac{3}{7}$

$\int_1^5 f(3x)\,dx = 7$

$= \frac{1}{3}\int_1^5 3f(3x)\,dx = \frac{1}{3}\int_3^{15} f(u)\,du = 7$

$u = 3x$
$du = 3dx$

$\int_3^{15} f(u)\,du = 21$

2. Choose the correct statement(s) given that $\int_0^9 f(x)\,dx = 11$ and $\int_3^9 f(x)\,dx = -1$.

✓ **I.** $\int_9^3 f(x)\,dx = 1$ ✗ **II.** $\int_3^0 f(x)\,dx = 6$ ✓ **III.** $\int_0^3 f(x)\,dx = 12$

(A) I only (B) II only (C) III only (D) I and II $\boxed{\text{(E) I and III}}$

$\int_0^3 f(x)dx + \int_3^9 f(x)dx = \int_0^9 f(x)dx$

$\int_0^3 f(x)dx + -1 = 11$

$\int_0^3 f(x)dx = 12$

3. If the substitution $u = x - 1$ is made, then the integral $\int_2^5 x\sqrt{x-1}\,dx =$

(A) $\int_2^5 (u+1)\sqrt{u}\,du$ $\boxed{\text{(B) } \int_1^4 (u+1)\sqrt{u}\,du}$

(C) $\int_1^4 u\sqrt{u-1}\,du$ (D) $\int_2^5 u\sqrt{u-1}\,du$

$u = x - 1$
$du = dx$
$u(5) = 4$
$u(2) = 1$
$u + 1 = x$

4. $\lim_{n\to\infty} \sum_{k=1}^{n} \left(\sin^2\left(1 + \frac{4k}{n}\right) + 3\right)\left(\frac{4}{n}\right) =$

(A) $\int_0^4 (\sin^2 x + 3)dx$ $\boxed{\text{(B) } \int_1^5 (\sin^2 x + 3)dx}$

(C) $4\int_0^1 (\sin^2(1+x) + 3)dx$ (D) $\int_0^1 (\sin^2(1+4x) + 3)dx$

(A) $\Delta x = \frac{4}{n}$ $x_k = 0 + \frac{4k}{n}$

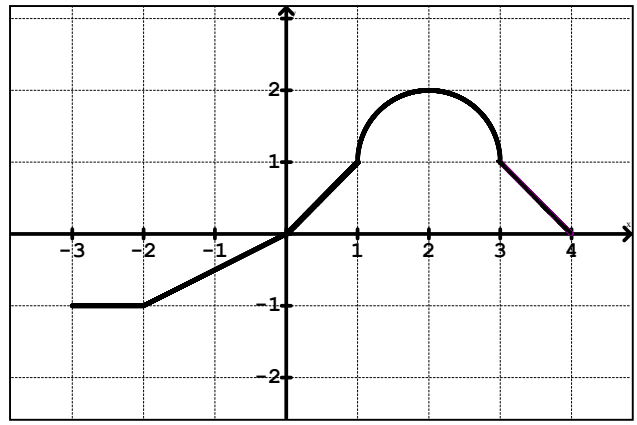
✓ (B) $\Delta x = \frac{5-1}{n} = \frac{4}{n}$ $x_k = 1 + \frac{4k}{n}$

(C) $\Delta x = \frac{1-0}{n} = \frac{1}{n}$ $x_k = 0 + \frac{1k}{n}$

(D) $\Delta x = \frac{1}{n}$ $x_k = 0 + \frac{1k}{n}$

$\left(\sin^2\left(1 + \frac{4k}{n}\right) + 3\right)\left(\frac{1}{n}\right)$

Use the graph of $f(x)$ below for problems **5** and **6.**



5. Find $\int_1^4 f(x)\,dx$.

$2+\frac{1}{2}\pi(1)^2$
$+\frac{1}{2}$
$=\frac{5}{2}+\frac{\pi}{2}$

$\boxed{\text{(A) } \frac{\pi+5}{2}}$ (B) $\frac{5}{2}+\pi$

(C) $\pi+3$ (D) $\frac{\pi}{2}+3$

6. Find $\int_{-3}^1 f(x)\,dx$.

$-1-\frac{1}{2}(2)(1)+\frac{1}{2}$
$-1-1+\frac{1}{2}$

(A) -2.5 $\boxed{\text{(B) } -1.5}$

(C) 3 (D) 2.5

7. If the substitution $u = \cos 3x$ is made, then the integral $\int_0^{\frac{\pi}{6}} \cos^3 3x \sin 3x\ dx =$

$u=\cos 3x$
$du=-3\sin 3x\,dx$

$\frac{-1}{3}\int_0^{\pi/6} -3(\cos 3x)^3 \sin 3x\,dx$
$=-\frac{1}{3}\int_1^0 u^3\,du$

(A) $-\frac{1}{3}\int_0^{\frac{\pi}{6}} u^3\,du$ (B) $\frac{1}{3}\int_0^{\pi} u^3\,du$

$\boxed{\text{(C) } -\frac{1}{3}\int_1^0 u^3\,du}$ (D) $-\frac{1}{3}\int_0^1 u^3\,du$

8. Consider $F(x) = \int_1^{2x} \frac{1}{1+t^3}\,dt$. Find $F'(x)$.

$F'(x)=\frac{1}{1+(2x)^3}\cdot 2$
$=\frac{2}{1+8x^3}$

(A) $\frac{1}{1+8x^2}$ (B) $\frac{1}{1+2x^2}$

(C) $\frac{2}{1+8x^3}-\frac{1}{2}$ $\boxed{\text{(D) } \frac{2}{1+8x^3}}$

Free Response Questions

1. Evaluate the integral.

$$\int \frac{16x^4 + 8x^2 + 3x}{x} dx$$

$= \int (16x^3 + 8x + 3)\,dx$

$= \boxed{4x^4 + 4x^2 + 3x + C}$

2. Evaluate the integral.

$$\int \left(30x^4 - \frac{14}{x^3}\right) dx$$

$= \int (30x^4 - 14x^{-3})\,dx$

$= 6x^5 + 7x^{-2} + C$

$= \boxed{6x^5 + \frac{7}{x^2} + C}$

3. Calculate the integral and write your answer in simplest form.

$$\int \frac{3}{\sqrt{x^5}} dx$$

$= \int 3x^{-5/2}\,dx$

$= 3 \cdot \frac{-2}{3} x^{-3/2} + C$

$= \boxed{-2x^{-3/2} + C}$

4. Calculate the integral and write your answer in simplest form.

$$\int \frac{3\sqrt[3]{x^5}}{2} dx$$

$= \int \frac{3}{2} x^{5/3}\,dx$

$= \frac{3}{2} \cdot \frac{3}{8} x^{8/3} + C$

$= \boxed{\frac{9}{16} x^{8/3} + C}$

5. Use u-substitution to evaluate the integral and express your answer in simplest form.

$$\int 4x^2 \sqrt[3]{4x^3 + 5}\, dx$$

$\frac{4}{12} \int 12x^2 \sqrt[3]{4x^3+5}\,dx$

$u = 4x^3 + 5$

$du = 12x^2\,dx$

$= \frac{1}{3} \int u^{1/2}\,du$

$= \frac{1}{3} \cdot \frac{2}{3} u^{3/2} + C = \boxed{\frac{2}{9}(4x^3+5)^{3/2} + C}$

6. Use u-substitution to evaluate the integral and express your answer in simplest form.

$$\int 3x(1 - 3x^2)^{\frac{3}{2}}\, dx$$

$u = 1 - 3x^2$

$du = -6x\,dx$

$\frac{3}{-6} \int -6x\,(1-3x^2)^{3/2}\,dx$

$= \frac{-1}{2} \int u^{3/2}\,du$

$= \frac{-1}{2} \cdot \frac{2}{5} u^{5/2} + C$

$= \boxed{\frac{-1}{5}(1-3x^2)^{5/2} + C}$

7. Use u-substitution to evaluate the integral and express your answer in simplest form.

$$\int \frac{2}{3\sqrt[3]{6x+1}}\,dx$$

$u = 6x+1$

$du = 6\,dx$

$\frac{1}{6}\cdot\frac{2}{3}\int \frac{1\cdot 6}{\sqrt[3]{6x+1}}\,dx$

$= \frac{1}{9}\int \frac{1}{\sqrt[3]{u}}\,du$

$= \frac{1}{9}\int u^{-1/3}\,du$

$= \frac{1}{9}\cdot\frac{3}{2}u^{2/3} + C$

$= \boxed{\frac{1}{6}(6x+1)^{2/3} + C}$

8. Use u-substitution to evaluate the integral and express your answer in simplest form.

$$\int \frac{18x}{(x^2+3)^4}\,dx$$

$u = x^2+3$

$du = 2x\,dx$

$\frac{18}{2}\int \frac{2x}{(x^2+3)^4}\,dx$

$= 9\int \frac{1}{u^4}\,du$

$= 9\int u^{-4}\,du$

$= 9\cdot\frac{-1}{3}u^{-3} + C$

$= -3(x^2+3)^{-3} + C$

$= \boxed{\frac{-3}{(x^2+3)^3} + C}$

9. Calculate the integral and express your answer in simplest form.

$$\int_0^1 \frac{3x^2\sqrt{9-5x^3}}{2}\,dx$$

$u = 9-5x^3$

$du = -15x^2\,dx$

$\int_0^1 \frac{3}{2}x^2\sqrt{9-5x^3}\,dx$

$= \frac{3}{2}\cdot\frac{-1}{15}\int_0^1 -15x^2\sqrt{9-5x^3}\,dx$

$= \frac{-1}{10}\int_9^4 \sqrt{u}\,du$

$= \frac{-1}{10}\int_9^4 u^{1/2}\,du$

$= \frac{-1}{10}\cdot\frac{2}{3}u^{3/2}\Big|_9^4$

$= \frac{-1}{15}u^{3/2}\Big|_9^4$

$= \frac{-1}{15}(8-27) = \frac{-1}{15}(-19) = \frac{19}{15}$

10. Calculate the integral and express your answer in simplest form.

$$\int_{-1}^{1} \frac{9x^2}{(5-x^3)^2}\,dx$$

$u = 5 - x^3$

$du = -3x^2\,dx$

$\frac{9}{-3}\int_{-1}^{1} \frac{-3x^2}{(5-x^3)^2}\,dx$

$= -3\int_{6}^{4} \frac{1}{u^2}\,du$

$= -3 \cdot \frac{-1}{u}\Big|_{6}^{4}$

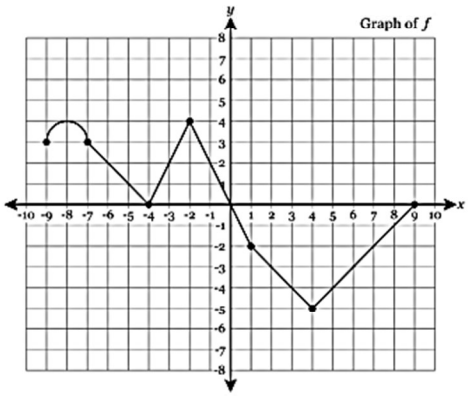
$= \frac{3}{u}\Big|_{6}^{4} = \frac{3}{4} - \frac{3}{6} = \frac{1}{4}$

11. Evaluate $\frac{d}{dx}\int_{3}^{5x} \cos(t^2)\,dt$

$= \cos((5x)^2) \cdot 5$

$= \boxed{5\cos(25x^2)}$

12. The graph of the function f in the figure below consists of line segments and a semicircle. Let g be the function given by $g(x) = \int_{-4}^{x} f(t)dt$. Determine all intervals on the closed interval $[-9, 9]$ where g is concave up.

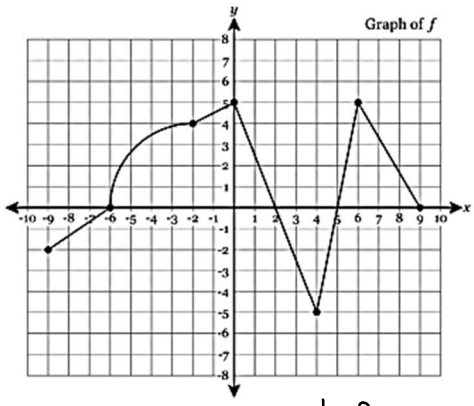


g is conc. up → $g'' > 0$ → g' inc.

$g' = f$ inc

$(-9,-8)\ (-4,-2)\ (4,9)$

13. The graph of the function f in the figure below consists of line segments and a quarter of a circle. Let g be the function given by $g(x) = \int_{5}^{x} f(t)dt$. Determine all values of x, if any, where g has a relative maximum on the open interval $(-9, 9)$.



rel max occurs where $g' = f$ changes from pos. to neg.

$\boxed{x = 2}$

14. Given the function $f(x) = \int_1^x \left(t^3 - 18t^2 + 81t\right) dt$, determine all x-values, if any, for which f has a point of inflection.

$f'' = 0$ and changes signs

$f'(x) = x^3 - 18x^2 + 81x$

$f''(x) = 3x^2 - 36x + 81$

$3(x^2 - 12x + 27) = 0$

$3(x-3)(x-9) = 0$

$+ \quad 3 \quad - \quad 9 \quad +$

$\boxed{x=3 \quad x=9}$

15. Evaluate the integral and express your answer in simplest form.

$$\frac{1}{2}\int \frac{2 \cdot 2}{\sqrt{25 - 16x^2}} dx$$

$u^2 = 16x^2$

$u = 4x \qquad a^2 = 25 \to a = 5$

$du = 4dx$

$$\boxed{\frac{1}{2} \cdot \arcsin\left(\frac{4x}{5}\right) + C}$$

16. Evaluate the integral and express your answer in simplest form.

$$\int \frac{-8}{1 + 16x^2} dx$$

$u^2 = 16x^2$

$u = 4x \qquad a^2 = 1 \to a = 1$

$du = 4dx$

$$-\frac{8}{4}\int \frac{1 \cdot 4}{1 + 16x^2} dx$$

$= -2 \cdot 1 \arctan(4x) + C$

$= \boxed{-2\arctan(4x) + C}$

17. Let f be a continuous function such that $\int_2^{10} f(x)\, dx = -3$ and $\int_{-1}^{10} f(x)\, dx = 3$. What is the value of $\int_{-1}^{2} \left(f(x) + 1\right) dx$?

$\int_{-1}^{2} f(x)dx + \int_{-1}^{2} 1 dx$

$= 6 + x\Big|_{-1}^{2}$

$= 6 + (2 - -1)$

$= 6 + 3$

$= \boxed{9}$

$\int_{-1}^{2} f(x)dx + \int_{2}^{10} f(x)dx = \int_{-1}^{10} f(x)dx$

$\int_{-1}^{2} f(x)dx + -3 = 3$

$\int_{-1}^{2} f(x)dx = 6$

18. Let f be a continuous function such that $\int_1^8 f(x)\, dx = 2$ and $\int_3^1 f(x)\, dx = 3$. What is the value of $\int_3^8 \left(f(x) + 5\right) dx$?

$\int_1^3 f(x)dx = -3$

$\int_1^3 f(x)dx + \int_3^8 f(x)dx = \int_1^8 f(x)dx$

$-3 + \int_3^8 f(x)dx = 2$

$\int_3^8 f(x)dx = 5$

$\int_3^8 (f(x) + 5)dx = \int_3^8 f(x)dx + \int_3^8 5\, dx$

$= 5 + \left(5x\Big|_3^8\right)$

$= 5 + 40 - 15 = \boxed{30}$